

06 Power-Weight Boundary for Rolling Validation

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Weighted rolling validation in the Gaussian mean example

The weighted simulations in Analysis 05 suggested that exponents near zero behave differently from larger positive or negative exponents. Here I study this boundary analytically in the Gaussian mean example.

Let $Y_1, \dots, Y_n$ be independent with $Y_i \sim N(\theta_*, 1)$. At validation time i , consider the sample-mean predictor

$$\hat{\theta}_{i-1} = \frac{1}{i-1} \sum_{t=1}^{i-1} Y_t$$

and the oracle predictor θ_* . For a fixed exponent $\alpha \in \mathbb{R}$, assign validation time i the weight $w_i = i^\alpha$. Writing $\xi_i = Y_i - \theta_*$, so that $\xi_i \stackrel{\text{iid}}{\sim} N(0, 1)$, the weighted rolling-validation loss difference is

$$\Xi_n^{(\alpha)} = \sum_{i=2}^n i^\alpha \left\{ (\hat{\theta}_{i-1} - \theta_*)^2 - 2\xi_i(\hat{\theta}_{i-1} - \theta_*) \right\}.$$

Positive values favor the oracle; $\Xi_n^{(\alpha)} < 0$ is the event that rolling validation incorrectly prefers the fitted sample-mean rule. The quantity studied below is

$$R_n(\alpha) = \frac{\text{Var}(\Xi_n^{(\alpha)})}{\{\mathbb{E}[\Xi_n^{(\alpha)}]\}^2}.$$

Since the mean is positive, Chebyshev's inequality gives

$$\mathbb{P}(\Xi_n^{(\alpha)} < 0) \leq \mathbb{P}\left(|\Xi_n^{(\alpha)} - \mathbb{E}\Xi_n^{(\alpha)}| \geq \mathbb{E}\Xi_n^{(\alpha)}\right) \leq R_n(\alpha).$$

Thus $R_n(\alpha) \rightarrow 0$ is sufficient for selection consistency. The converse does not follow from this bound.

Mean and variance decomposition

Put $S_m = \sum_{t=1}^m \xi_t$. Then $\hat{\theta}_{i-1} - \theta_\star = S_{i-1}/(i-1)$, which is independent of the centered variable ξ_i . Consequently,

$$\mathbb{E} \left[-2\xi_i \frac{S_{i-1}}{i-1} \right] = 0, \quad \mathbb{E} \left[\frac{S_{i-1}^2}{(i-1)^2} \right] = \frac{1}{i-1},$$

and hence

$$\mathbb{E}[\Xi_n^{(\alpha)}] = \sum_{i=2}^n \frac{i^\alpha}{i-1}. \quad (1)$$

Because $i^\alpha/(i-1) \sim i^{\alpha-1}$, the series comparison and integral test yield

$$\begin{aligned} \mathbb{E}[\Xi_n^{(\alpha)}] &\rightarrow M_\infty(\alpha), & \alpha < 0, \\ \mathbb{E}[\Xi_n^{(0)}] &\sim \log n, & \alpha = 0, \\ \mathbb{E}[\Xi_n^{(\alpha)}] &\sim n^\alpha/\alpha, & \alpha > 0. \end{aligned} \quad (2)$$

where $M_\infty(\alpha) = \sum_{i=2}^\infty i^\alpha/(i-1) \in (0, \infty)$ for $\alpha < 0$.

To analyze the variance, expand each S_{i-1}^2 , collect the coefficients of ξ_t^2 , and then collect the coefficient of each product $\xi_k \xi_l$, $k < l$. Define

$$a_{n,t} = \sum_{i=t+1}^n \frac{i^\alpha}{(i-1)^2}, \quad 1 \leq t \leq n-1,$$

and

$$b_{n,l} = \sum_{i=l+1}^n \frac{i^\alpha}{(i-1)^2} - \frac{l^\alpha}{l-1}, \quad 2 \leq l \leq n.$$

The first sum in $b_{n,l}$ comes from the squared running means at validation times $i > l$, while the second term comes from the cross-product at validation time l . Rearranging the finite sums gives

$$\Xi_n^{(\alpha)} = \sum_{t=1}^{n-1} a_{n,t} \xi_t^2 + 2 \sum_{l=2}^n b_{n,l} \sum_{k=1}^{l-1} \xi_k \xi_l. \quad (3)$$

After centering the diagonal part, its summands are multiples of $\xi_t^2 - 1$. A diagonal term and an off-diagonal product have zero covariance because at least one independent centered Gaussian factor appears to the first power. The same observation shows that two distinct products $\xi_k \xi_l$ and $\xi_r \xi_s$ are orthogonal unless their unordered index pairs agree. Finally, $\text{Var}(\xi_t^2) = 2$ and $\mathbb{E}(\xi_k^2 \xi_l^2) = 1$ for $k \neq l$. It follows from (3) that

$$\text{Var}(\Xi_n^{(\alpha)}) = 2 \sum_{t=1}^{n-1} a_{n,t}^2 + 4 \sum_{l=2}^n (l-1) b_{n,l}^2. \quad (4)$$

Asymptotic behavior under power weights

Proposition. For each fixed $\alpha \in \mathbb{R}$,

$$R_n(\alpha) \rightarrow \begin{cases} R_\infty(\alpha) > 0, & \alpha < 0, \\ 0, & \alpha = 0, \\ \frac{2\alpha^2}{1+\alpha}, & \alpha > 0. \end{cases} \quad (5)$$

Proof. Suppose first that $\alpha < 0$. Since $i^\alpha/(i-1)^2 \leq C i^{\alpha-2}$, comparison with the corresponding integral gives, uniformly in n ,

$$0 \leq a_{n,t} \leq C t^{\alpha-1}.$$

The triangle inequality gives the same order for the off-diagonal coefficients,

$$|b_{n,l}| \leq \sum_{i=l+1}^n \frac{i^\alpha}{(i-1)^2} + \frac{l^\alpha}{l-1} \leq C l^{\alpha-1}.$$

The dominating series $\sum_t t^{2\alpha-2}$ and $\sum_l l^{2\alpha-1}$ converge because $2\alpha-2 < -1$ and $2\alpha-1 < -1$. For each fixed index the coefficients converge as $n \rightarrow \infty$, so dominated convergence applied to (4) yields a finite limit $V_\infty(\alpha)$. This limit is positive because the limiting diagonal coefficients are positive. Together with (2), this proves

$$R_n(\alpha) \rightarrow \frac{V_\infty(\alpha)}{M_\infty(\alpha)^2} > 0. \quad (6)$$

For $\alpha = 0$, $a_{n,t} \leq \sum_{k=t}^{\infty} k^{-2} \leq C/t$, so the diagonal sum in (4) is bounded. For the off-diagonal coefficient, write

$$b_{n,l} = \sum_{k=l}^{n-1} \frac{1}{k^2} - \frac{1}{l-1}.$$

The integral bounds for the infinite tail give

$$\left| \sum_{k=l}^{\infty} k^{-2} - \frac{1}{l-1} \right| \leq Cl^{-2}, \quad \sum_{k=n}^{\infty} k^{-2} \leq Cn^{-1}.$$

Therefore $|b_{n,l}| \leq C(l^{-2} + n^{-1})$. Using $(x+y)^2 \leq 2x^2 + 2y^2$,

$$\sum_{l=2}^n (l-1)b_{n,l}^2 \leq C \sum_{l=2}^n l^{-3} + Cn^{-2} \sum_{l=2}^n l = O(1).$$

Equation (4) now gives $\text{Var}(\Xi_n^{(0)}) = O(1)$, while (1) equals the harmonic sum $H_{n-1} \sim \log n$. Hence $R_n(0) = O((\log n)^{-2}) \rightarrow 0$.

It remains to consider $\alpha > 0$. The diagonal coefficients satisfy

$$a_{n,t} \leq C \sum_{i=t+1}^n i^{\alpha-2}.$$

For $0 < \alpha < 1$, this is at most $Ct^{\alpha-1}$, and therefore

$$\sum_{t=1}^{n-1} a_{n,t}^2 = \begin{cases} O(1), & 0 < \alpha < 1/2, \\ O(\log n), & \alpha = 1/2, \\ O(n^{2\alpha-1}), & 1/2 < \alpha < 1. \end{cases}$$

For $\alpha = 1$, the same sum is $O(n)$, since $a_{n,t} \leq C\{1 + \log(n/t)\}$ and $n^{-1} \sum_{t \leq n} \{1 + \log(n/t)\}^2$ is bounded by an integrable Riemann sum. For $\alpha > 1$, $a_{n,t} \leq Cn^{\alpha-1}$, giving $\sum_t a_{n,t}^2 = O(n^{2\alpha-1})$. Every one of these bounds is $o(n^{2\alpha})$, so the diagonal contribution is negligible on the required scale.

For the off-diagonal term, put $x = l/n$. On every interval $x \in [\varepsilon, 1]$, the integral approximation for the coefficient sum is uniform and gives

$$n^{1-\alpha} b_{n,l} \longrightarrow g_{\alpha}(x) = \int_x^1 u^{\alpha-2} du - x^{\alpha-1}. \quad (7)$$

When $\alpha \neq 1$, this function is

$$g_\alpha(x) = \frac{1 - \alpha x^{\alpha-1}}{\alpha - 1}.$$

For $0 < \alpha < 1$, the omitted endpoint requires care. The integral comparison used above also gives $|b_{n,l}| \leq Cl^{\alpha-1}$, and hence

$$n^{-2\alpha} \sum_{l \leq \varepsilon n} lb_{n,l}^2 \leq \frac{C}{n} \sum_{l \leq \varepsilon n} (l/n)^{2\alpha-1} \leq C\varepsilon^{2\alpha}. \quad (8)$$

Thus the small- x contribution vanishes after $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$; on $[\varepsilon, 1]$, (7) supplies the ordinary Riemann-sum limit. For $\alpha > 1$, the bound $|b_{n,l}| \leq Cn^{\alpha-1}$ makes the normalized contribution from $l \leq \varepsilon n$ at most $C\varepsilon^2$, while g_α is bounded at zero. Consequently,

$$n^{-2\alpha} \sum_{l=2}^n (l-1)b_{n,l}^2 \rightarrow \int_0^1 xg_\alpha(x)^2 dx. \quad (9)$$

For $\alpha \neq 1$, direct integration yields

$$\int_0^1 xg_\alpha(x)^2 dx = \frac{1}{(\alpha-1)^2} \left\{ \frac{1}{2} - \frac{2\alpha}{\alpha+1} + \frac{\alpha}{2} \right\} = \frac{1}{2(\alpha+1)}. \quad (10)$$

When $\alpha = 1$, (7) becomes $g_1(x) = \log(1/x) - 1$. Its logarithmic singularity is integrable against $x dx$, and truncation uses the bound $C\{1 + \log(n/l)\}$ exactly as for the diagonal term. Moreover,

$$\int_0^1 x\{\log(1/x) - 1\}^2 dx = \frac{1}{4},$$

which agrees with (10) at $\alpha = 1$. Combining (4), (9), (10), and the negligible diagonal contribution proves

$$\text{Var}(\Xi_n^{(\alpha)}) \sim \frac{2}{1+\alpha} n^{2\alpha}.$$

Dividing by the square of the mean n^α/α from (2) proves the positive- α line of (5). $\square$

Table 1: Asymptotic behavior for fixed power-weight exponents.

Regime	Mean	Variance	Ratio
$\alpha < 0$	$\rightarrow M_\infty(\alpha) \in (0, \infty)$	$\rightarrow V_\infty(\alpha) \in (0, \infty)$	$\rightarrow R_\infty(\alpha) > 0$
$\alpha = 0$	$\sim \log n$	$O(1)$	$\rightarrow 0$
$\alpha > 0$	$\sim n^\alpha/\alpha$	$\sim 2n^{2\alpha}/(1 + \alpha)$	$\rightarrow 2\alpha^2/(1 + \alpha)$

Critical value of α

Among fixed power weights, $R_n(\alpha) \rightarrow 0$ only at

$$\alpha^* = 0.$$

This is a point boundary, not an empirical cutoff such as 0.1 or 0.2. The positive-side limit $2\alpha^2/(1 + \alpha)$ approaches zero as $\alpha \downarrow 0$, and the negative-side constant can likewise be small near zero. Consequently, a small nonzero exponent can mimic a vanishing ratio over a finite range even though its fixed- α limit is positive.

A preliminary calculation suggested $R_n(\alpha) \rightarrow \alpha/2$ for $\alpha > 0$. Recomputing the exact off-diagonal coefficient gives instead $2\alpha^2/(1 + \alpha)$, as shown in (10) after applying the variance factor 4 and normalizing by $\{n^\alpha/\alpha\}^2$.

Numerical verification

The numerical calculations use the exact coefficient identity (4) and an independent Monte Carlo experiment. Lines in the figures are exact finite- n theory; points are empirical estimates. The simulations are diagnostic and do not enter the proof.

Table 2: Numerical settings for Analysis 06.

setting	value
Seed	2026
Monte Carlo replications	5000
Batch size	250
Sample sizes	100, 200, 500, 1000, 2000, 5000, 10000
Power exponents	-1, -0.5, -0.1, 0, 0.1, 0.25, 0.5, 1

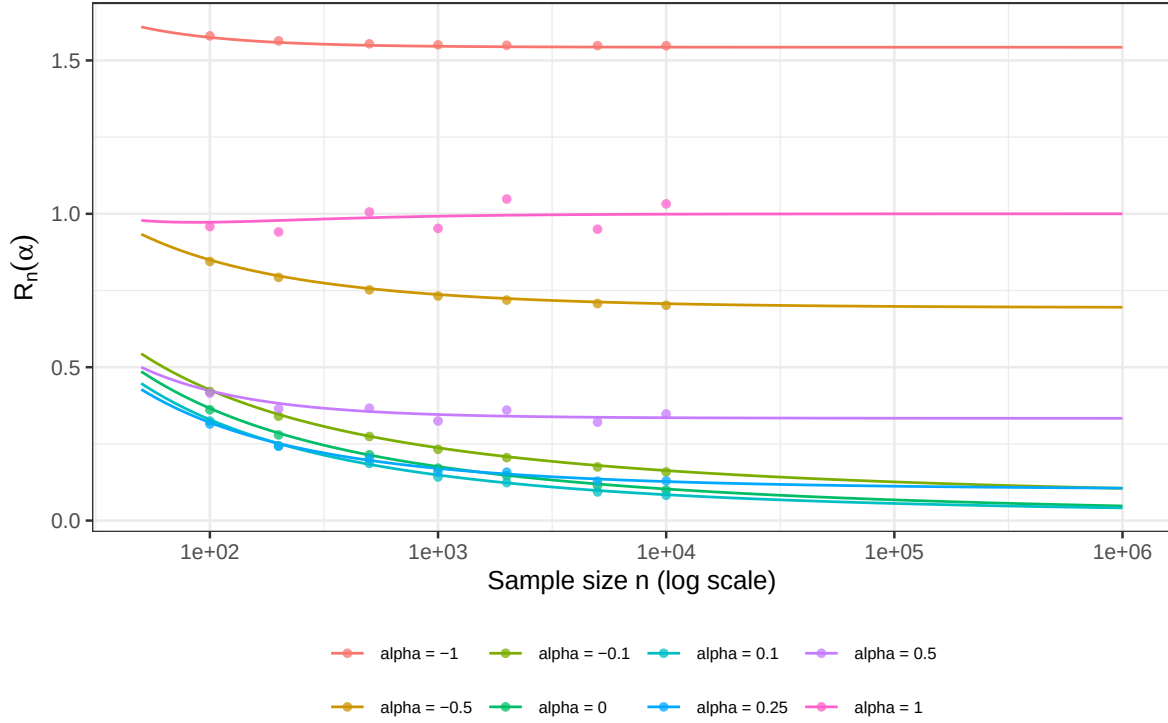


Figure 1: Exact Chebyshev-ratio trajectories (lines) and Monte Carlo variance estimates divided by the exact squared mean (points).

For small positive exponents convergence is slow: the finite- n curve can remain well above its limiting level even at large n . This is the finite-sample behavior that made the boundary difficult to identify from Analysis 05 alone.

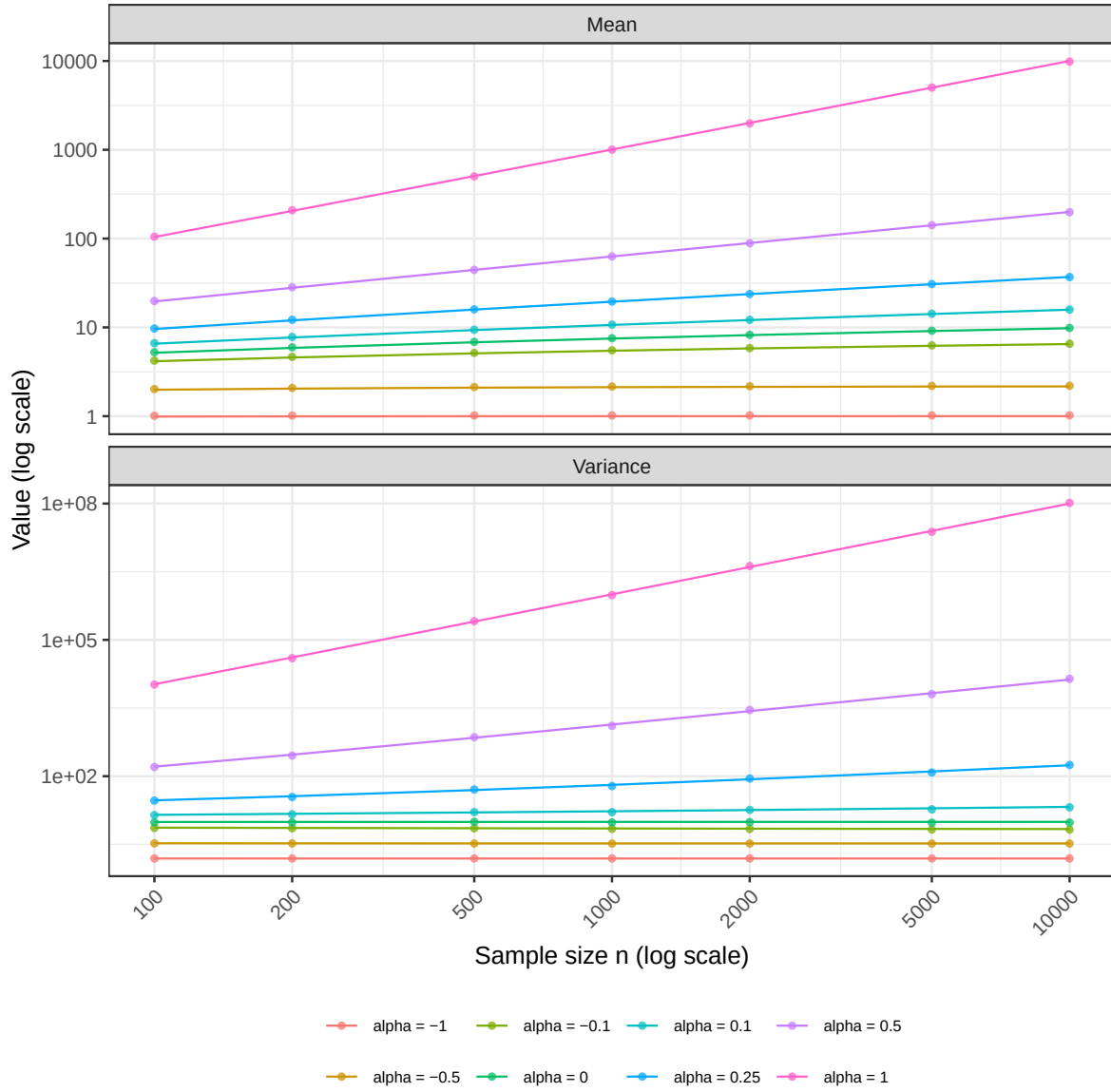


Figure 2: Exact means and variances (lines) compared with Monte Carlo estimates (points). Each panel uses a logarithmic vertical scale.

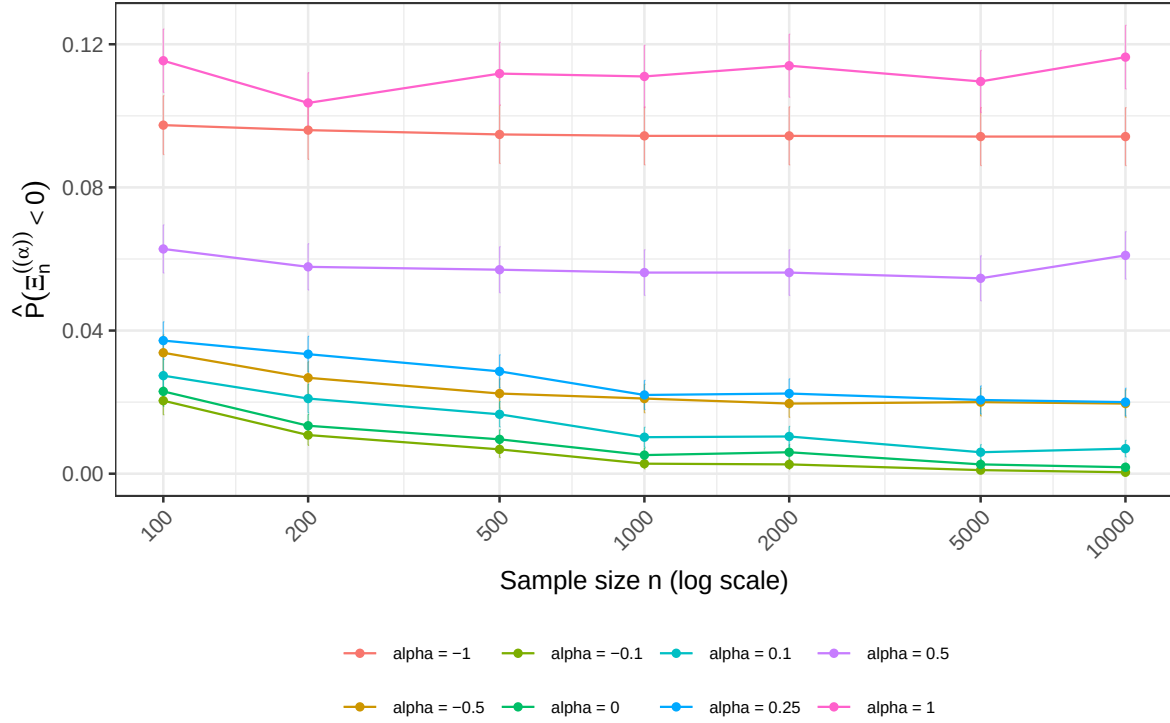


Figure 3: Estimated incorrect-selection probability, with pointwise 95% Monte Carlo error bars.

The final figure displays the limiting ratio. Negative-exponent points are numerical evaluations of the convergent infinite coefficient series; the nonnegative curve uses the closed-form result in (5).

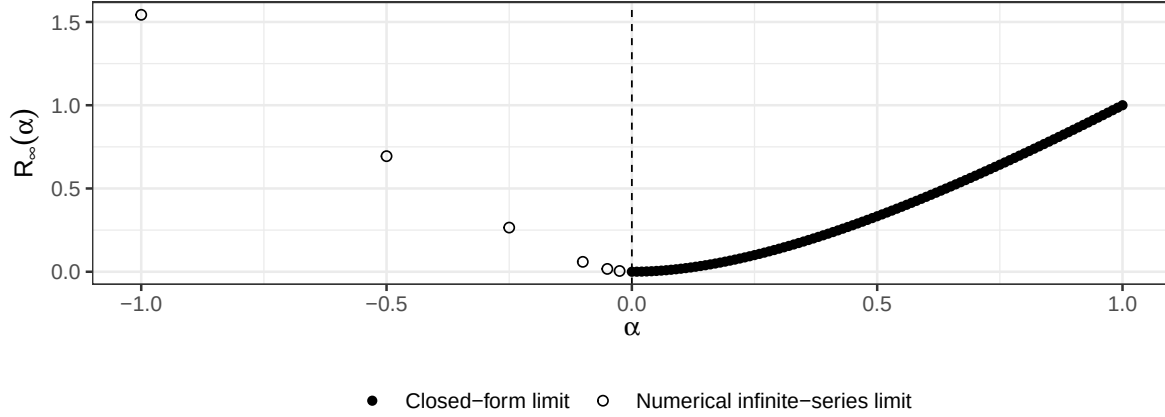


Figure 4: Asymptotic Chebyshev-ratio limit as a function of the fixed power exponent. The dashed line marks the unique zero at $\alpha = 0$.

The negative-side limits were numerically stable when the truncation level was increased from 5×10^5 to 10^6 .

Table 3: Truncation stability of negative-alpha ratio limits.

Alpha	Limit at 1,000,000	Limit at 500,000	Absolute change
-1.000	1.543017	1.543017	0
-0.500	0.694126	0.694126	0
-0.250	0.265202	0.265202	0
-0.100	0.059413	0.059413	0
-0.050	0.016973	0.016973	0
-0.025	0.004560	0.004560	0

Interpretation of the result

The effect of the weights depends on how they change both the mean and the variance of the rolling-validation score. When $\alpha < 0$, later observations receive less weight, and both the mean and variance approach finite positive limits. When $\alpha = 0$, the mean grows like $\log n$, while cancellation in the variance calculation keeps the variance bounded. Therefore, $R_n(0) \rightarrow 0$, and Chebyshev's inequality shows that the probability of selecting the wrong model tends to zero.

When $\alpha > 0$, the mean grows faster, but the variance grows at the same order as the squared mean. As a result,

$$R_n(\alpha) \longrightarrow \frac{2\alpha^2}{1+\alpha} > 0.$$

Thus, a faster-growing mean does not necessarily lead to a vanishing ratio. Among fixed power exponents, $\alpha = 0$ is the only value for which $R_n(\alpha) \rightarrow 0$.

This conclusion concerns the ratio $R_n(\alpha)$. For $\alpha \neq 0$, its positive limit means that Chebyshev's inequality does not establish selection consistency; it does not prove inconsistency either. Determining whether the actual incorrect-selection probability tends to zero would require a further argument.